信 息 工 程 学 院

《大数据采集与清洗》实验报告

学院：信息工程学院　**班级**：数科 2102　**姓名**：陈　**学号**:202101　**成绩**:

一、实验题目

使用 Flume 接收数据

二、实验目的

使用 Flume 工具接收来自 AvroSource 和 NetcatSource 的数据

三、实验平台

希冀平台

Ubuntu

Flume 1.7.0

四、实验过程及步骤

实验 1 接收 AvroSource 的信息

1、创建 Agent 配置文件

```
a1.sources = r1
a1.sinks = k1
a1.channels = c1
a1.sources.r1.type = avro
a1.sources.r1.channels = c1
a1.sources.r1.bind = localhost
a1.sources.r1.port = 4141
a1.sinks.k1.type = logger
a1.channels.c1.type = memory
a1.channels.c1.capacity = 1000
a1.channels.c1.transactionCapacity = 100
a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1
~
~
~
~
~
~
~
~
~
"avro.conf" 14L, 340C                                    1,1
```

实验报告的内容与格式按任课教师的要求书写。

2、启动 Flume Agent al

```
efaultSinkProcessor@12bdc2d3 counterGroup:{ name:null counters:{} } }} channels:
{c1=org.apache.flume.channel.MemoryChannel{name: c1}} }
23/11/29 08:09:31 INFO node.Application: Starting Channel c1
23/11/29 08:09:31 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:09:31 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:09:31 INFO node.Application: Starting Sink k1
23/11/29 08:09:31 INFO node.Application: Starting Source r1
23/11/29 08:09:31 INFO source.AvroSource: Starting Avro source r1: { bindAddress
: localhost, port: 4141 }...
23/11/29 08:09:31 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: r1: Successfully registered new MBean.
23/11/29 08:09:31 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: r1 started
23/11/29 08:09:31 INFO source.AvroSource: Avro source r1 started.
```

3、创建指定文件

```
root@cg:~# mkdir -p /course/da/flume_op
root@cg:~# cd /course/da/flume_op/
root@cg:/course/da/flume_op# touch log.00
root@cg:/course/da/flume_op# echo hello world > log.00
root@cg:/course/da/flume_op# cat log.00
hello world
```

4、成功接收信息

```
ted.
23/11/29 08:13:20 INFO sink.LoggerSink: Event: { headers:{} body: 68 65 6C 6C 6F
 20 77 6F 72 6C 64               hello world }
```

实验 2 接收 NetcatSource 的信息

1、创建 Agent 配置文件

```
a1.sources = r1
a1.sinks = k1
a1.channels = c1
a1.sources.r1.type = netcat
a1.sources.r1.bind = localhost
a1.sources.r1.port = 44444
a1.sinks.k1.type = logger
a1.channels.c1.type = memory
a1.channels.c1.capacity = 1000
a1.channels.c1.transactionCapacity = 100l
a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1
~
```

2、启动 Flume Agent al

```
ied, initializing channel to default capacity of 100
23/11/29 08:23:49 INFO node.AbstractConfigurationProvider: Created channel c1
23/11/29 08:23:49 INFO source.DefaultSourceFactory: Creating instance of source
r1, type netcat
23/11/29 08:23:49 INFO sink.DefaultSinkFactory: Creating instance of sink: k1, t
ype: logger
23/11/29 08:23:49 INFO node.AbstractConfigurationProvider: Channel c1 connected
to [r1, k1]
23/11/29 08:23:49 INFO node.Application: Starting new configuration:{ sourceRunn
ers:{r1=EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource{n
ame:r1,state:IDLE} }} sinkRunners:{k1=SinkRunner: { policy:org.apache.flume.sink
.DefaultSinkProcessor@70bb792e counterGroup:{ name:null counters:{} } }} channel
s:{c1=org.apache.flume.channel.MemoryChannel{name: c1}} }
23/11/29 08:23:49 INFO node.Application: Starting Channel c1
23/11/29 08:23:49 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:23:49 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:23:49 INFO node.Application: Starting Sink k1
23/11/29 08:23:49 INFO node.Application: Starting Source r1
23/11/29 08:23:49 INFO source.NetcatSource: Source starting
23/11/29 08:23:49 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.Serv
erSocketChannelImpl[/127.0.0.1:44444]
```

3、安装 telnet 服务

```
root@cg:~# apt-get install telnet
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  telnet
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 63.5 kB of archives.
After this operation, 182 kB of additional disk space will be used.
Get:1 http://mirrors.aliyun.com/ubuntu xenial/main amd64 telnet amd64 0.17-40 [6
3.5 kB]
Fetched 63.5 kB in 0s (196 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package telnet.
(Reading database ... 80574 files and directories currently installed.)
Preparing to unpack .../telnet_0.17-40_amd64.deb ...
Unpacking telnet (0.17-40) ...
Processing triggers for man-db (2.7.5-1) ...
Setting up telnet (0.17-40) ...
update-alternatives: using /usr/bin/telnet.netkit to provide /usr/bin/telnet (te
lnet) in auto mode
root@cg:~#
```

4、传递字符

```
root@cg:~# telnet localhost 44444
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
hello world
OK
```

```
s:{c1=org.apache.flume.channel.MemoryChannel{name: c1}} }
23/11/29 08:23:49 INFO node.Application: Starting Channel c1
23/11/29 08:23:49 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:23:49 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:23:49 INFO node.Application: Starting Sink k1
23/11/29 08:23:49 INFO node.Application: Starting Source r1
23/11/29 08:23:49 INFO source.NetcatSource: Source starting
23/11/29 08:23:49 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.Serv
erSocketChannelImpl[/127.0.0.1:44444]
23/11/29 08:25:09 INFO sink.LoggerSink: Event: { headers:{} body: 68 65 6C 6C 6F
 20 77 6F 72 6C 64 0D             hello world. }
```

实验 3 监听目录中新增文件

1、创建 Agent 配置文件

```
a1.sources = r1
a1.sinks = k1
a1.channels = c1

a1.sources.r1.type = spooldir
a1.sources.r1.spoolDir = /course/da/flume_op/lookup
a1.sources.r1.fileSuffix = .ok
a1.sources.r1.deletePolicy = never
a1.sources.r1.fileHeader = true

a1.sinks.k1.type = logger

a1.channels.c1.type = memory
a1.channels.c1.capacity = 1000
a1.channels.c1.transactionCapacity = 100

a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1
```

```
root@cg:/usr/local/flume/conf# vim dir.conf
root@cg:/usr/local/flume/conf# mkdir /course/da/flume_op/lookup
root@cg:/usr/local/flume/conf#
```

2、启动 Flume Agent

```
r1, type spooldir
23/11/29 08:29:31 INFO sink.DefaultSinkFactory: Creating instance of sink: k1, t
ype: logger
23/11/29 08:29:31 INFO node.AbstractConfigurationProvider: Channel c1 connected
to [r1, k1]
23/11/29 08:29:31 INFO node.Application: Starting new configuration:{ sourceRunn
ers:{r1=EventDrivenSourceRunner: { source:Spool Directory source r1: { spoolDir:
 /course/da/flume_op/lookup } }} sinkRunners:{k1=SinkRunner: { policy:org.apache
.flume.sink.DefaultSinkProcessor@3626babb counterGroup:{ name:null counters:{} }
 }} channels:{c1=org.apache.flume.channel.MemoryChannel{name: c1}} }
23/11/29 08:29:31 INFO node.Application: Starting Channel c1
23/11/29 08:29:31 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:29:31 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:29:31 INFO node.Application: Starting Sink k1
23/11/29 08:29:31 INFO node.Application: Starting Source r1
23/11/29 08:29:31 INFO source.SpoolDirectorySource: SpoolDirectorySource source
starting with directory: /course/da/flume_op/lookup
23/11/29 08:29:31 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: r1: Successfully registered new MBean.
23/11/29 08:29:31 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: r1 started
```

3、创建文件

```
Terminal
File  Edit  View  Terminal  Tabs  Help
hello you
hello me
hello him
Genshin, start!
~
~
~
~
~
~
~
~
```

```
23/11/29 08:31:05 INFO sink.LoggerSink: Event: { headers:{file=/course/da/flume_
op/lookup/test.txt} body: 68 65 6C 6C 6F 20 79 6F 75                      hello
you }
23/11/29 08:31:05 INFO sink.LoggerSink: Event: { headers:{file=/course/da/flume_
op/lookup/test.txt} body: 68 65 6C 6C 6F 20 6D 65                         hello
me }
23/11/29 08:31:05 INFO sink.LoggerSink: Event: { headers:{file=/course/da/flume_
op/lookup/test.txt} body: 68 65 6C 6C 6F 20 68 69 6D                      hello
him }
23/11/29 08:31:05 INFO sink.LoggerSink: Event: { headers:{file=/course/da/flume_
op/lookup/test.txt} body: 47 65 6E 73 68 69 6E 2C 20 73 74 61 72 74 21    Genshi
n, start! }
```

```
root@cg:~# ls /course/da/flume_op/lookup/
test.txt.ok
```

实验 4 监听文件中新增数据

1、创建 Agent 配置文件

```
a1.sources = r1
a1.sinks = k1
a1.channels = c1

a1.sources.r1.type = exec
a1.sources.r1.command = tail -F /course/da/flume_op/lookup/data.log

a1.sinks.k1.type = logger

a1.channels.c1.type = memory
a1.channels.c1.capacity = 10000000
a1.channels.c1.transactionCapacity = 1000000

a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1

~
~
~
~
~
~
~
"data.conf" 16L, 334C                                    1,1
```

```
root@cg:/usr/local/flume/conf# touch /course/da/flume_op/lookup/data.log
root@cg:/usr/local/flume/conf#
```

2、启动 Flume Agent

```
efaultSinkProcessor@633cb9dc counterGroup:{ name:null counters:{} } }} channels:
{c1=org.apache.flume.channel.MemoryChannel{name: c1}} }
23/11/29 08:35:05 INFO node.Application: Starting Channel c1
23/11/29 08:35:05 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:35:05 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:35:05 INFO node.Application: Starting Sink k1
23/11/29 08:35:05 INFO node.Application: Starting Source r1
23/11/29 08:35:05 INFO source.ExecSource: Exec source starting with command:tail
 -F /course/da/flume_op/lookup/data.log
23/11/29 08:35:05 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: r1: Successfully registered new MBean.
23/11/29 08:35:05 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: r1 started
```

3、添加数据

```
root@cg:~# echo Genshin, start! > /course/da/flume_op/lookup/data.log
root@cg:~#
```

```
group for type: SOURCE, name: r1: Successfully registered new MBean.
23/11/29 08:35:05 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: r1 started
23/11/29 08:35:59 INFO sink.LoggerSink: Event: { headers:{} body: 47 65 6E 73 68
 69 6E 2C 20 73 74 61 72 74 21    Genshin, start! }
```

实验 5 向 HDFS 存储数据

1、创建 Agent 配置文件

```
a1.sinks.k1.hdfs.fileSuffix = .log
a1.sinks.k1.hdfs.inUsePrefix = xttzm.

a1.sinks.k1.hdfs.inUseSuffix = .zdhm
a1.sinks.k1.hdfs.rollInterval = 0
a1.sinks.k1.hdfs.rollSize = 0
a1.sinks.k1.hdfs.rollCount = 5
a1.sinks.k1.hdfs.useLocalTimeStamp = true

a1.sinks.k1.hdfs.fileType = DataStream

a1.sinks.k1.hdfs.writeFormat = Text

a1.sinks.k1.hdfs.minBlockReplicas = 1

a1.channels.c1.type = file
a1.channels.c1.checkpointDir = /course/da/flume_op/lookup/checkpoint
a1.channels.c1.transactionCapacity = 1000000
a1.channels.c1.dataDirs = /course/da/flume_op/lookup/data

a1.sources.r1.channels = c1
a1.sinks.k1.channel = c1

-- INSERT --                                                    34,1
```

```
root@cg:/usr/local/flume/conf# hadoop fs -mkdir -p /course/flume
root@cg:/usr/local/flume/conf# mkdir -p /course/da/flume_op/lookup/checkpoint
root@cg:/usr/local/flume/conf# mkdir -p /course/da/flume_op/lookup/data
root@cg:/usr/local/flume/conf#
```

2、启动 Flume Agent

```
urse/da/flume_op/lookup/checkpoint/checkpoint, elements to sync = 0
23/11/29 08:43:11 INFO file.EventQueueBackingStoreFile: Updating checkpoint meta
data: logWriteOrderID: 1701247391879, queueSize: 0, queueHead: 0
23/11/29 08:43:12 INFO file.Log: Updated checkpoint for file: /course/da/flume_o
p/lookup/data/log-1 position: 0 logWriteOrderID: 1701247391879
23/11/29 08:43:12 INFO file.FileChannel: Queue Size after replay: 0 [channel=c1]
23/11/29 08:43:12 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: CHANNEL, name: c1: Successfully registered new MBean.
23/11/29 08:43:12 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/11/29 08:43:12 INFO node.Application: Starting Sink k1
23/11/29 08:43:12 INFO node.Application: Starting Source r1
23/11/29 08:43:12 INFO source.NetcatSource: Source starting
23/11/29 08:43:12 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SINK, name: k1: Successfully registered new MBean.
23/11/29 08:43:12 INFO instrumentation.MonitoredCounterGroup: Component type: SI
NK, name: k1 started
23/11/29 08:43:12 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.Serv
erSocketChannelImpl[/127.0.0.1:52019]
```

3、传递字符

```
root@cg:~# telnet localhost 52019
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
hello you
OK
hello me
OK
Genshin, start!
OK
```

```
23/11/29 08:43:12 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.Serv
erSocketChannelImpl[/127.0.0.1:52019]
23/11/29 08:44:11 INFO file.EventQueueBackingStoreFile: Start checkpoint for /co
urse/da/flume_op/lookup/checkpoint/checkpoint, elements to sync = 1
23/11/29 08:44:12 INFO file.EventQueueBackingStoreFile: Updating checkpoint meta
data: logWriteOrderID: 1701247391882, queueSize: 1, queueHead: 999999
23/11/29 08:44:12 INFO file.Log: Updated checkpoint for file: /course/da/flume_o
p/lookup/data/log-1 position: 90 logWriteOrderID: 1701247391882
23/11/29 08:44:12 INFO hdfs.HDFSDataStream: Serializer = TEXT, UseRawLocalFileSy
stem = false
23/11/29 08:44:12 INFO hdfs.BucketWriter: Creating hdfs://master:9000/course/flu
me//xttzm.http.1701247452187.log.zdhm
23/11/29 08:44:42 INFO file.EventQueueBackingStoreFile: Start checkpoint for /co
urse/da/flume_op/lookup/checkpoint/checkpoint, elements to sync = 1
23/11/29 08:44:42 INFO file.EventQueueBackingStoreFile: Updating checkpoint meta
data: logWriteOrderID: 1701247391892, queueSize: 0, queueHead: 0
23/11/29 08:44:42 INFO file.Log: Updated checkpoint for file: /course/da/flume_o
p/lookup/data/log-1 position: 470 logWriteOrderID: 1701247391892
```

```
root@cg:~# hadoop dfs -cat /course/flume//xttzm.http.1701247452187.log.zdhm
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.

hello you
hello me
Genshin, start!
```

实验作业

实验代码

```
#! /bin/bash

function listenFromAvro(){
    `flume-ng agent -c /usr/local/flume/conf -f /usr/local/flume/conf/avro.conf -n a1
-Dflume.root.logger=INFO,console`
}

function listenFromNetcat(){
```

```
    `flume-ng agent -c /usr/local/flume/conf/ -f /usr/local/flume/conf/example.conf --n a1 -Dflume.root.logger=INFO,console`
}

function listenFromDir(){
    echo -e "\n\n"
    read -p "Please Input the Directory Path: " dirPath

    if [ ! -d $dirPath ];
    then
        echo "The Directory Does Not Exist!"
        exit 1
    fi

    sed -i "s:a1.sources.r1.spoolDir =.*:a1.sources.r1.spoolDir = $dirPath:g" /usr/local/flume/conf/dir.conf

    `flume-ng agent -c /usr/local/flume/conf -f /usr/local/flume/conf/dir.conf -n a1 -Dflume.root.logger=INFO,console`
}

function listenFromFile(){
    echo -e "\n\n"
    read -p "Please Input the File Path: " filePath

    if [ ! -e $filePath ];
    then
        echo "The File Does Not Exist!"
        exit 1
    fi

    sed -i "a1.sources.r1.command = tail -F.*:a1.sources.r1.command = tail -F $filePath:g" /usr/local/flume/conf/data.conf

    `flume-ng agent -c /usr/local/flume/conf -f /usr/local/flume/conf/data.conf -n a1 -Dflume.root.logger=INFO,console`
}

function writeToHDFS(){
    `flume-ng agent -c /usr/local/flume/conf -f /usr/local/flume/conf/hdfs.conf -n a1 -Dflume.root.logger=INFO,console`
}

echo "-------------------------------"
```

```
echo "Welcome to This System"
echo "Input 1~5 For:"
echo -e "\t1: Listen From AvroSource"
echo -e "\t2: Listen From NetcatSource"
echo -e "\t3: Listen From Directory"
echo -e "\t4: Listen From File"
echo -e "\t5: Write To HDFS"
echo "-------------------------------"
read -p "Please Enter Your Choice: " choice

case "${choice}" in
    "1")
        listenFromAvro
        ;;
    "2")
        listenFromNetcat
        ;;
    "3")
        listenFromDir
        ;;
    "4")
        listenFromFile
        ;;
    "5")
        writeToHDFS
        ;;
    *)
        echo "Your Input Is Invalid!"
        ;;
esac
```

接收 AvroSource 的信息

```
root@cg:/usr/local/flume# ./exp.sh
-------------------------------
Welcome to This System
Input 1~5 For:
        1: Listen From AvroSource
        2: Listen From NetcatSource
        3: Listen From Directory
        4: Listen From File
        5: Write To HDFS
-------------------------------
Please Enter Your Choice: 1
Info: Including Hadoop libraries found via (/opt/module/hadoop-2.7.1/bin/hadoop)
 for HDFS access
Info: Including HBASE libraries found via (/opt/module/hbase-1.1.5/bin/hbase) fo
r HBASE access
Info: Including Hive libraries found via (/opt/module/hive) for Hive access
```

```
7.0.0.1:4141] BOUND: /127.0.0.1:4141
23/12/07 06:34:40 INFO ipc.NettyServer: [id: 0xac04262e, /127.0.0.1:49012 => /12
7.0.0.1:4141] CONNECTED: /127.0.0.1:49012
23/12/07 06:34:41 INFO ipc.NettyServer: [id: 0xac04262e, /127.0.0.1:49012 :> /12
7.0.0.1:4141] DISCONNECTED
23/12/07 06:34:41 INFO ipc.NettyServer: [id: 0xac04262e, /127.0.0.1:49012 :> /12
7.0.0.1:4141] UNBOUND
23/12/07 06:34:41 INFO ipc.NettyServer: [id: 0xac04262e, /127.0.0.1:49012 :> /12
7.0.0.1:4141] CLOSED
23/12/07 06:34:41 INFO ipc.NettyServer: Connection to /127.0.0.1:49012 disconnec
ted.
23/12/07 06:34:44 INFO sink.LoggerSink: Event: { headers:{} body: 68 65 6C 6C 6F
 20 77 6F 72 6C 64                hello world }
```

接收 **NetcatSource** 的信息

```
root@cg:/usr/local/flume# ./exp.sh
-------------------------------
Welcome to This System
Input 1~5 For:
        1: Listen From AvroSource
        2: Listen From NetcatSource
        3: Listen From Directory
        4: Listen From File
        5: Write To HDFS
-------------------------------
Please Enter Your Choice: 2
Info: Including Hadoop libraries found via (/opt/module/hadoop-2.7.1/bin/hadoop)
 for HDFS access
Info: Including HBASE libraries found via (/opt/module/hbase-1.1.5/bin/hbase) fo
r HBASE access
```

```
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
genshin start
OK
```

```
23/12/07 06:35:44 INFO node.Application: Starting Sink k1
23/12/07 06:35:44 INFO node.Application: Starting Source r1
23/12/07 06:35:44 INFO source.NetcatSource: Source starting
23/12/07 06:35:44 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.Serv
erSocketChannelImpl[/127.0.0.1:44444]
23/12/07 06:36:30 INFO sink.LoggerSink: Event: { headers:{} body: 67 65 6E 73 68
 69 6E 20 73 74 61 72 74 0D       genshin start. }
```

监听目录中新增文件

```
root@cg:/usr/local/flume# ./exp.sh
-------------------------------
Welcome to This System
Input 1~5 For:
        1: Listen From AvroSource
        2: Listen From NetcatSource
        3: Listen From Directory
        4: Listen From File
        5: Write To HDFS
-------------------------------
Please Enter Your Choice: 3



Please Input the Directory Path: /course/da/flume_op/lookup
Info: Including Hadoop libraries found via (/opt/module/hadoop-2.7.1/bin/hadoop
 for HDFS access
Info: Including HBASE libraries found via (/opt/module/hbase-1.1.5/bin/hbase) f
```

```
root@cg:/usr/local/flume# touch /course/flume_op/lookup/f1.txt
touch: cannot touch '/course/flume_op/lookup/f1.txt': No such file or directory
root@cg:/usr/local/flume# touch /course/da/flume_op/lookup/f1.txt
root@cg:/usr/local/flume# echo genshin start > /course/da/flume_op/lookup/f1.txt
root@cg:/usr/local/flume#
```

```
run(ScheduledThreadPoolExecutor.java:294)
        at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.
java:1149)
        at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor
.java:624)
        at java.lang.Thread.run(Thread.java:748)
23/12/07 06:39:12 INFO sink.LoggerSink: Event: { headers:{file=/course/da/flume_
op/lookup/f1.txt} body: 67 65 6E 73 68 69 6E 20 73 74 61 72 74          genshin
start }
```

监听文件中新增数据

```
root@cg:/usr/local/flume# touch /course/da/flume_op/lookup/data2.log
root@cg:/usr/local/flume# ./exp.sh
-------------------------------
Welcome to This System
Input 1~5 For:
        1: Listen From AvroSource
        2: Listen From NetcatSource
        3: Listen From Directory
        4: Listen From File
        5: Write To HDFS
-------------------------------
Please Enter Your Choice: 4



Please Input the File Path: /course/da/flume_op/lookup/data2.log
Info: Including Hadoop libraries found via (/opt/module/hadoop-2.7.1/bin/hadoo
 for HDFS access
```

```
root@cg:/usr/local/flume# echo Genshin start > /course/da/flume_op/lookup/data2.
log
root@cg:/usr/local/flume#
```

```
23/12/07 06:50:42 INFO instrumentation.MonitoredCounterGroup: Component type: CH
ANNEL, name: c1 started
23/12/07 06:50:42 INFO node.Application: Starting Sink k1
23/12/07 06:50:42 INFO node.Application: Starting Source r1
23/12/07 06:50:42 INFO source.ExecSource: Exec source starting with command:tail
 -F /course/da/flume_op/lookup/data2.log
23/12/07 06:50:42 INFO instrumentation.MonitoredCounterGroup: Monitored counter
group for type: SOURCE, name: r1: Successfully registered new MBean.
23/12/07 06:50:42 INFO instrumentation.MonitoredCounterGroup: Component type: SO
URCE, name: r1 started
23/12/07 06:50:46 INFO sink.LoggerSink: Event: { headers:{} body: 47 65 6E 73 68
 69 6E 20 73 74 61 72 74          Genshin start }
```